Ahmed Abdelmonem Mohamed Abdlgayoum

Mechatronics Engineer · Embedded Systems, Robotics & Automation

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PROFESSIONAL SUMMARY

Mechatronics Engineering graduate who builds systems end to end, from SolidWorks design and embedded C/C++ firmware to sensor integration. Focused on embedded systems and robotics, including real-time computer vision on low-power edge devices, backed by industrial experience in electrical testing, instrumentation and PLC process control. Seeking an entry-level engineering role in embedded systems, robotics or automation.

EXPERIENCE

Electrical & Instrumentation Engineer Jul 2025 – Oct 2025

Hi Technics System Sdn Bhd, Cyberjaya, Malaysia

- Carried out panel wiring, electrical testing and instrumentation work on industrial systems in line with industrial safety standards.
- Performed insulation resistance and earth continuity testing to verify installations before energisation.
- Troubleshot electrical and instrumentation faults on site to reduce equipment downtime.
- Assisted with field sensor installation and prepared supporting technical documentation.

PROJECTS

Real-Time Embedded Vision System for Autonomous Navigation

Raspberry Pi 5 · YOLOv11 · ONNX · NCNN · Python · OpenCV

- Built an on-device traffic-sign detection system for an autonomous vehicle that runs entirely on a Raspberry Pi 5 CPU, with no GPU or cloud.
- Exported a YOLOv11 model to ONNX and optimised it with the NCNN framework for real-time edge inference.
- Linked detection output to decision logic and motor control, closing the loop from perception to actuation.

Smart Autonomous Lawn Mower Robot

ESP32 · Raspberry Pi · YOLOv8 · MobileNetV2 · OpenCV · Python · DC motors

- Designed a two-processor architecture: ESP32 for real-time motor control and Raspberry Pi for vision processing.
- Ran a comparative study of YOLOv8 and MobileNetV2 for grass detection to select the model best suited to accuracy and on-board navigation.

Autonomous Pickup & Waste-Sorting Robot (ROS2 Simulation)

ROS2 · Gazebo · Ubuntu · YOLOv8 · LiDAR · SolidWorks

- Modelled a mobile robot with a 6-DOF arm in SolidWorks and simulated it in Gazebo for object pickup and waste sorting.
- Integrated YOLOv8 object detection with camera and LiDAR inputs within a ROS2 architecture.

Automated Soap Production System (Hardware Prototype)

Arduino Mega · ESP32 · Level sensors · DHT11 · Pumps · Motors

- Developed a multi-stage process controller coordinating pumps, motors and temperature regulation across Arduino Mega and ESP32.
- Implemented level sensing and sequenced process stages, mirroring industrial batch-control logic on physical hardware.

PLC Process Automation: Filling Tank & Warehouse Storage (Factory I/O)

PLC · Ladder logic · Factory I/O

- Programmed ladder logic for an automated tank-filling process: sequential pump control, liquid filling and level detection.
- Designed an automated storage-and-retrieval system controlling conveyors and sensors through sequential control logic.

SKILLS

Programming	C, C++, Embedded C, Python
Embedded hardware	Raspberry Pi 5, ESP32, ESP8266, STM32, Arduino
Robotics & vision	ROS2, Gazebo, OpenCV, YOLO (v8/v11), ONNX, NCNN
Automation & E&I	PLC / ladder logic, Factory I/O, panel wiring, insulation & continuity testing
CAD / EDA	SolidWorks, Fusion 360, Proteus, EasyEDA

EDUCATION

Bachelor of Mechatronics Engineering (Hons.) Oct 2022 – Jul 2026

Universiti Malaysia Pahang Al-Sultan Abdullah (UMPSA)

CERTIFICATIONS & LANGUAGES

The Complete Electronics Course: Analog Hardware Design, Udemy (Jan 2026)

Languages: Arabic (native) · English (fluent) · German (basic)